What Makes for a Good Saliency Map? Comparing Strategies for Evaluating Saliency Maps in Explainable AI (XAI)

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Saliency maps are a popular approach for explaining classifications of (convolutional) neural networks. However, it remains an open question as to how best to evaluate salience maps, with three families of evaluation methods commonly being used: subjective user measures, objective user measures, and mathematical metrics. We examine three of the most popular saliency map approaches (viz., LIME, Grad-CAM, and Guided Backpropagation) in a between subject study (N=166) across these families of evaluation methods. We test 1) for subjective measures, if the maps differ with respect to user trust and satisfaction; 2) for objective measures, if the maps increase users' abilities and thus understanding of a model; 3) for mathematical metrics, which map achieves the best ratings across metrics; and 4) whether the mathematical metrics can be associated with objective user measures. To our knowledge, our study is the first to compare several salience maps across all these evaluation methods—with the finding that they do not agree in their assessment (i.e., there was no difference concerning trust and satisfaction, Grad-CAM improved users' abilities best, and Guided Backpropagation had the most favorable mathematical metrics). Additionally, we show that some mathematical metrics were associated with user understanding, although this relationship was often counterintuitive. We discuss these findings in light of general debates concerning the complementary use of user studies and mathematical metrics in the evaluation of explainable AI (XAI) approaches.

CCS Concepts: • **Human-centered computing → User studies**; *Empirical studies in HCI*; • **Social and professional topics →** *Socio-technical systems*; Computing literacy.

Additional Key Words and Phrases: saliency maps, user study, functionally-grounded evaluation, user abilities, user satisfaction, user trust, image classification, transparency, explainable AI, XAI

1 Introduction

The use of saliency maps to evaluate image classifiers, often based on convolutional neural networks (CNNs), has emerged as a significant area of research within the field of explainable AI (XAI). Saliency maps are popular explanation tools that provide visual representations highlighting which parts of an image are most influential in the decision-making process of a CNN, offering a window into the otherwise opaque workings of these models. This research is particularly important for applications where trust in and understanding of AI decisions are crucial (e.g., medical imaging [7, 66, 67]): by making the decision-making process of CNNs more transparent, saliency maps can empower users to exert better control and oversight, thereby aligning with regulatory requirements such as those set out by the EU AI Act [24].

Enhancing user desiderata (e.g., satisfaction or understanding) is a key goal of XAI approaches like saliency maps [17, 46]. Achieving this goal requires consistent evaluation and comparison of different approaches. One popular family of evaluation methods is user studies [5, 47, 70], where approaches are evaluated either subjectively through questionnaires [35, 36], or objectively, using user abilities or task performance [58, 79]. A more formalized family of evaluation methods focuses on certain properties (e.g., fidelity) that can be assessed via mathematical metrics [15, 30, 44].

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When it comes to the evaluation of XAI approaches, some researchers view mathematical metrics as a complement to user studies [23], whereas others hope that these metrics could partially even replace user studies [8, 9, 16]. The reason for this hope is that user studies have several limitations. For example, they can be subject to individual biases (like change blindness or a preference for usability over performance [11]) and are resource intensive, and thus often not scalable. In contrast, mathematical metrics are objective and easy to scale. They can be applied across large datasets and diverse models, allowing for widespread benchmarking of XAI approaches. However, mathematical metrics also have their disadvantages. For example, different metrics for the same property can diverge [85], leading to contradicting evaluations. Furthermore, user studies are indispensable for capturing human perspectives on desiderata such as trust and satisfaction, and for evaluating whether XAI approaches really lead to better understanding and better abilities in tasks where AI models are used.

Although there are many studies that have evaluated saliency maps with user studies *or* mathematical metrics, these studies often assess only one type of saliency map [58] or focus on one type of evaluation method [85]. Without explicitly comparing different saliency maps using evaluation methods across families, we may miss to understand which ones are better suited to satisfy specific user desiderata. Furthermore, whether mathematical metrics only complement or can partially even replace user studies has only been assessed to a limited extent and for singular metrics.

Against this backdrop, we aim to contribute to research on evaluating XAI approaches by comparing several popular types of saliency maps using representatives of each family of evaluation methods. At the same time, we aim to shed more light on the relationship between user studies and mathematical metrics more generally. To this end, we conducted a between-subjects online experiment where participants were tasked to, among other things, estimate the accuracy of an image classifier based on saliency maps that were generated using either Gradient-Weighted Class Activation Mapping (Grad-CAM) [76], Guided Backpropagation (GBP) [80], or Local Interpretable Model-Agnostic Explanations (LIME) [70]. We see three main contributions of this paper:

(1) We compare three saliency map approaches regarding their effects on user reactions, user abilities, and regarding their mathematical metrics. We do so in a task that is close to real-life application scenarios of saliency maps (i.e., using them to enhance understanding and assess the performance of image classifiers).
(2) We show that mathematical metrics and user abilities seem to be complementary evaluation methods: the approach that was evaluated most favorable with respect to the mathematical metrics (in this study GBP) was not the one that was most useful when it comes to enhancing user abilities (in this study Grad-CAM).
(3) We show that whereas some mathematical metrics can be associated with user abilities (e.g., as proxies), these associations can be counterintuitive and difficult to interpret. Together with (2), this calls into question the assumption that mathematical metrics can replace user studies.

2 Background and Related Work

2.1 AI Image Classification and Saliency Maps

Image classification is a task where AI models categorize images into predefined classes. It is widely used in fields such as healthcare (e.g., for diagnosing medical images) [7, 66, 67], security (e.g., for facial recognition) [29, 68], autonomous vehicles (e.g., for object detection) [51], and retail (e.g., for visual search engines) [63]. Research on image classification includes improving model accuracy [69], augmenting adversarial robustness [22], and reducing bias [53].

With the use of image classification in sensitive areas such as healthcare, the issue of trustworthiness has become increasingly important [49]. XAI aims to contribute to assessing the trustworthiness of AI-based systems [39]. This involves, for example, finding out whether different types of explanations can enable users to better understand classifiers or better assess which outputs are correct and which are incorrect [50]. Likewise, it also involves empowering users to make better decisions based on AI outputs [10]. Furthermore, XAI can also contribute to more favorable reactions towards image classifiers as people may find them more helpful for their everyday goals and work tasks [64].

A popular type of explanation for AI image classification are saliency maps [78]. They highlight the regions of an image that are most influential in the AI model's decision-making process, providing a visual representation of what the AI "sees" as important. In general, saliency maps aim to make the decision process of AI systems more transparent, allowing users to better understand the model and make informed decisions when it should and should not be trusted [70]. At the same time, saliency maps can also help developers to identify and fix mistakes in AI models [26], thus providing a tool for laypersons and experts alike. In the context of this paper, we will highlight three popular methods of saliency mapping: Grad-CAM [76] (≈23,500 citations), GBP [80] (≈6,000 citations), and LIME [70] (≈23,000 citations).[1]

Each of the methods for generating saliency maps works differently and has its own visual properties. For instance, Grad-CAM visualizes the *areas* in the image that contribute most to the predicted class by leveraging the gradients flowing into the last convolutional layer. Visually, Grad-CAM highlights areas it deems important using different colors, for example, by using warmer colors for high important areas and colder colors for areas of low importance. LIME perturbs the input data and observes changes in the output to identify which parts of the image are most important. This results in individual *sections* being separated and highlighted. Finally, GBP modifies the standard backpropagation process, allowing only positive gradients to contribute to the visualization, resulting in more focused feature maps. These saliency maps look like they highlight mostly the *edges* of the most important features used for classification.

2.2 Evaluating XAI Approaches

When it comes to evaluating XAI approaches, a distinction is commonly made between two families of methods: human-centered evaluations (i.e., studies) and functionally grounded evaluations (i.e., mathematical metrics) [3, 8, 23, 82, 87]. For the former, there are two further sub-families that sometimes overlap [82, 87]: studies using subjective measures (e.g., with questionnaires) and studies using objective measures (e.g., with task performance metrics). In this section, we will briefly introduce each of these three (sub-)families.

In addition to evaluations that directly examine XAI approaches, there are also indirect evaluation methods, especially for saliency maps, which investigate whether, for example, the highlighted areas of a saliency map correspond to those that a person would use to categorize an image (see, e.g., [20, 27, 38, 45, 56, 57, 77]). While such evaluations are also typically part of studies, they are not within the scope of our research as they are rather meant to influence the design of future methods.

2.2.1 User Studies with Subjective Measures. When it comes to user studies that use subjective measures to evaluate different XAI approaches, trust and explanation satisfaction are often mentioned as important user reactions [14, 35, 36]. Trust can be defined as "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party" [52, p. 712]. Explanation satisfaction combines "the degree to which users feel that they understand the AI system or process being explained to them" [35, p. 3] and their "feeling of satisfaction" [35, p. 4] with the explanation.

[1]Citations according to Google Scholar (date of retrieval: 20th of April, 2025).

Findings on the impact of XAI approaches on user reactions such as trust and satisfaction are mixed. A recent review by Rong et al. [71] showed that only about half of the studies comparing explanations to no explanations or placebo (randomly generated) explanations found a positive effect on trust. In the context of saliency maps, comparative studies have yielded insights but also left open questions. For instance, Khadivpour et al. [40] and Ihongbe et al. [37] report that participants tend to prefer Grad-CAM over LIME, perceiving it as more understandable or trusting it more. However, trust and understandability in these studies are mostly measured by using single-item measures (e.g., asking participants to rank their preference [40]). While single-item measures can be an efficient choice to indicate rough inclinations in terms of preferences, they provide limited insights into potentially multifaceted constructs such as trust [4, 75]. To our knowledge, there also exists no investigation into user perceptions of GBP. Through our study, we want to address these gaps in research and provide a more rigorous investigation of different methods in terms of user-perceived satisfaction and trust. Against this backdrop, we ask the following open research questions (RQs)[2]:

RQ1.1 Are there differences in perceived explanation satisfaction between the three approaches?

RQ1.2 Are there differences in trust in the classifier and in the saliency map between the three approaches?

2.2.2 User Studies with Objective Measures. Increasing user understanding through explanations is a key goal of XAI research [17]. For example, explanations are crucial to enable users to understand and exert control over systems that could otherwise cause harm [10]. Through increased understanding, humans can stay in control of high-stakes decision scenarios involving automation [1]. This ensures that humans remain being the ones accountable for the outcomes systems might produce, something that is important enough to be explicitly mentioned by legislation such as the EU AI Act [24].

Building on Speith et al. [79], we believe that a promising way to grasp whether different explanations enhance user understanding is to examine their impact on certain user abilities interacting with AI. For saliency maps as explanations, important and often measured abilities related to how users interpret the results, also called the *perceptibility* of saliency maps [13], are how accurately users can predict AI accuracy and AI predictions [58]. This is because increased understanding of a system should enable users to predict its output [5, 59, 79]. A way to measure this is to ask users what percentage of images would be classified correctly by an image classifier for a specific class of images, or by presenting images and asking which of them would be classified correctly by the image classifier.

Research on the effectiveness of saliency maps in enhancing user understanding is mostly focused on Grad-CAM, with mixed results and methodological limitations. For instance, Colin et al. [19] investigated whether Grad-CAM's can improve user prediction accuracy across three classifiers, observing gains in one classifier but not the others. Similarly, Khadivpour et al. [40] examined both Grad-CAM and LIME, revealing similar accuracy levels but using a small sample size of 20 participants with only five predictions per condition, limiting the statistical power to detect meaningful differences. In contrast, Morrison et al. [55] found that Grad-CAM outperformed LIME in a game-based setting. Finally, Kim et al. [42] demonstrated that Grad-CAM enabled above-chance-level predictions. However, they did not statistically compare user performance across different saliency map approaches.

Though we already know some things about the influence of different saliency map approaches on user understanding, the question regarding whether one approach achieves consistently better results over others remains unclear. Our work not only fills the gap on the degree to which GBP might enable users to better understand classifiers but also

[2]The experiment was preregistered before data collection, the preregistration can be viewed under https://aspredicted.org/YT9_DXG. In the preregistration, we formulated the hypotheses that GBP would be preferred over the other two approaches both in terms of user reactions and abilities. We have based this thesis on considerations relating to human perception, which, according to our initial research, prefers edges over surfaces. However, we realized that the literature on the relevance of edges vs. surfaces is more diverse than we initially though. This is why we now propose open RQs.

provides another crucial piece in the puzzle by systematically comparing Grad-CAM, LIME, and GBP using a robust sample size and diverse evaluation methods. Because the literature shows no clear benefit of one approach over the others, we propose the following open RQs[3]:

RQ2.1 Are there differences in participants' accuracy prediction relative to the true accuracy between the three approaches?

RQ2.2 Are there differences in the number of participants' classification errors between the three approaches?

2.2.3 Mathematical Metrics. Besides evaluation through studies with humans, mathematical metrics play a key role in evaluating XAI approaches [15, 48]. In the field of XAI, mathematical metrics for evaluating saliency maps are primarily used to evaluate properties such as *fidelity* [30, 31], *robustness* [21, 74], *complexity* [12, 61], and *localization* [48].

Fidelity, also known as faithfulness or correctness, describes an explanation's ability to accurately reflect the predictions of a black-box model [16]. Fidelity is a crucial concept for an explanation since it is meant to state whether an explanation reproduces the dynamics of the underlying model [6]. In our user study, we ask humans to predict the model based on saliency maps, thus we expect this property to be particularly important. *Robustness*, or sensitivity, expresses the consistency of explanations under input perturbations. Since users may struggle to understand how a classifier works when presented with inconsistent explanations for slightly varying inputs [18], we expect they would prefer saliency maps with lower sensitivity. *Complexity* expresses the conciseness of explanations, meaning that only a few features are needed to explain a model's prediction. We expect that concise explanations may be easier to understand because they focus on a few key regions rather than emphasizing everything. Finally, for image explanations, *localization* portrays how well saliency maps align with human-annotated bounding boxes indicating object locations in an image. We expect a strong correlation between localization and human evaluations, as identifying targeted objects within an image aligns with the process humans use to classify images [86].

With various mathematical metrics designed to assess specific properties, key questions arise regarding their effectiveness in reflecting saliency map performance. For instance, while crucial in computer vision, *localization* has been criticized for its inability to differentiate between errors in the saliency map and the model [48]. More generally, Zhang et al. [85] and Li et al. [48] have conducted extensive experiments across metrics, primarily focusing on benchmarking. Both teams observed that no single method outperforms others across all metrics. While Grad-CAM performed fairly well across most metrics, metrics for the same property sometimes diverged when comparing different saliency maps. Against the background of these inconclusive findings, we thus ask:

RQ3 Are there differences in mathematical metrics that measure the properties fidelity, robustness, complexity, and localization between the three approaches?

While improving mathematical metrics offers undeniable benefits, their relationship to human understanding remains underexplored. Although saliency methods are designed to provide explanations that aid in understanding how models operate, there is often a disconnect between user expectations and the actual effectiveness of these explanations.

There are some results concerning the question of whether mathematical metrics can be associated (e.g., as proxies) with user studies. Nguyen et al. [62] and Kim et al. [42], for instance, found that metrics associated with localization do not indicate how well humans can judge the correctness of predictions with the help of different saliency maps such as Grad-CAM. Similarly, Colin et al. [19] found that metrics associated with fidelity are not associated with what they

[3]We also measured the number of features that participants thought the classifier relied on based on the saliency map in each trial (following Alqaraawi et al. [5]). However, because it is difficult to differentiate relevant from irrelevant features and because the number of detected features does not necessarily correspond to better understanding, we do not report these results. Results show that participants in the GBP group detected more features than participants in the LIME group, with no differences for the other comparisons. Results can be made available on request.

call the “utility” of a saliency map (i.e., how well participants could anticipate a classifier’s prediction based on diverse saliency maps including Grad-CAM). We aim to expand these findings by broadening the scope of metrics and saliency map approaches. This is why we examine multiple metrics representing different properties (i.e., fidelity, complexity, robustness, and localization). To our knowledge, this is also the first study to investigate LIME and GBP in this context.

We thus ask[4] :

RQ4 Are the mathematical metrics that measure the properties fidelity, robustness, complexity, and localization associated with how well participants predict the accuracy of different classes?

3 Methods

3.1 Design and Sample

For our one-factor (saliency map: Grad-CAM vs. GBP vs. LIME) between-subjects design we conducted an a priori power analysis for a one-way ANOVA and three groups using *G*Power* [25]. According to the analysis, a sample of 174 participants was needed to detect a small to medium effect size of f = .175, with a power of $1 - \beta$ = .80 and an α of .05.

We implemented our questionnaire using *SoSci Survey* and recruited English-speaking graduates in information or communication technology (ICT) via *Prolific*. We decided for this sample because we needed participants to develop a reasonable level of understanding of saliency maps and expected ICT graduates to have a higher affinity for the topic than the general population. ICT graduates may (in the future) also be the ones who assess the quality of AI models. The study was approved by the Ethical Review Board of the university of one of the first authors.

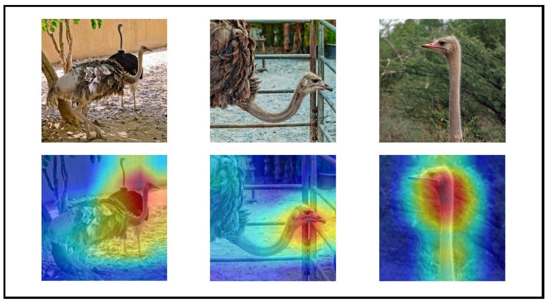
We recruited 175 participants who completed all questions, each receiving £6 to participate in the study. The average completion time was 28.03 minutes with a standard deviation (SD) of 9.21 minutes, leading to an average hourly wage of £12.84. Of the participants, we excluded 1 because they indicated that their data should not be used, and 6 who indicated that they did not at least have a bachelor’s or equivalent degree. The final sample consisted of 168 participants (28.6% female and 1.2% diverse; 4.8% with PhD or equivalent, 27.98% with a master’s degree or equivalent; M_{age} = 31.19, SD_{age} = 7.83). Roughly half (51.79%) of the participants reported no experience with saliency maps, 44.64% reported “a little” or “some” experience, and 3.57% reported “a lot” of experience. Almost all participants (96.43%) either “somewhat” or “strongly” agreed that they developed a good understanding of how to use saliency maps to evaluate AI classifiers after reading the explanations on saliency maps and completing the two example trials, while 1.19% “strongly” or “mostly” disagreed and 2.38% answered “neutral.”

3.2 Procedure

Participants were randomly assigned to one of three groups, corresponding to the different saliency map approaches Grad-CAM, GBP, or LIME. They first read a text explaining what saliency maps are (see Section A.3) followed by a text explaining how to interpret AI classifications using the respective saliency map approach of the group they were assigned to (see Section A.3.1, Section A.3.2, and Section A.3.3). Afterwards, participants were asked three comprehension questions, each had 4 answer options of which only one was correct (see Section A.3.4). If they answered one question incorrectly, they were transferred back to the explanation texts. Participants who answered incorrectly a second time were automatically screened out of the experiment.

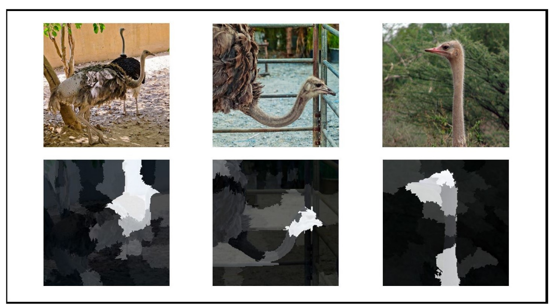
[4]Because subjective user metrics are prone to errors (such as the illusion of understanding [54]), we chose to focus on examining the relation of mathematical metrics to more objective measures of user understanding, i.e., participants’ accuracy predictions, instead.

After answering the comprehension questions correctly, participants were shown two example trials followed by 12 experimental trials. In each trial, 3 saliency maps were displayed in addition to the 3 original images from which the saliency maps were generated (see Figure 1). In the example trials, all the tasks that later needed to be performed by participants in each trial were explained. These tasks were to 1) estimate the accuracy of the classifier for the respective class, and 2) identify which of the six images from the specific class, shown alongside saliency maps, would be correctly classified by the image classifier (see Section A.3.5).



(a) Grad-CAM

(b) Guided Backpropagation



(c) LIME

Fig. 1. Example trial images from the three groups.

After the practice trials, participants indicated how confident they felt that they understood how to evaluate a classifier using saliency maps and subsequently completed the 12 experimental trials. Lastly, participants answered questions regarding their perceived explanation satisfaction, trust in the saliency maps, trust in and trustworthiness of the classifier, demographic questions, and whether their data can be used.

3.3 Measures and Materials

3.3.1 Network and Dataset. We use the pretrained ResNet50 [32], a classic CNN model, from the Pytorch model zoo[5]. We utilize sampled images from the ImageNet ILSVRC 2012 validation set, a widely used dataset renowned for its diversity and large scale, providing a standardized benchmark for image classification evaluation [43, 73], comprising 50,000 images evenly distributed across 1,000 categories. We randomly chose 12 classes from these 1,000. We had three

[5]https://pytorch.org/vision/0.8/models.html

images per trial for the saliency maps and six images per class that needed to be classified by participants, so we had 9×12 images for the experimental trials and 2×9 for the practice trials.

3.3.2 Saliency maps. The implementation of Grad-CAM [76] and GuidedBackprop [80] is from Gildenblat and contributors's GitHub package [28] while the implementation of LIME [70] is from its officially released GitHub [70].

3.3.3 Mathematical Metrics. We selected Insertion [65], Deletion [65], and Monotonicity [8] as representative metrics for fidelity; Average Sensitivity (Avg.Sensitivity) for robustness; Entropy Complexity (Ent.Complexity) and Sparseness as two metrics for complexity; and Energy Pointing Game (EPG) for localization. The implementation of Insertion and Deletion is from its official GitHub [65]. The implementation of the other metrics is is from the Quantus GitHub [33].

Concerning fidelity, *Insertion* and *Deletion* [65] sequentially add or remove pixels in decreasing order of saliency and observe the effect on the prediction. Deletion measures the decrease in the probability of a class when pixels are removed, with removal being equivalent to setting pixel values to zero (i.e., black). Conversely, Insertion measures the increase in probability when pixels are gradually reintroduced to a blurred version of the image (created by applying a Gaussian blur to the original image). Similar to Insertion, *Monotonicity* [8] measures the increase in classification probability when inserting pixels. However, instead of directly evaluating the probability, it assesses the correlation between the probability and the saliency map values as pixels are inserted. High values in Insertion and Monotonicity as well as a low value in Deletion indicate that the saliency map captures the most important parts of the input.

As a metric for robustness, *Avg. Sensitivity* [84] measures the average change in the saliency map under a small general perturbation. We give a spare of radius around the input to bound the perturbation, then calculate the average difference of saliency map values between the original input and perturbed input. A higher average sensitivity indicates a greater influence of input changes on the saliency map.

With respect to the complexity of the saliency maps, *Ent.Complexity* [12] evaluates the entropy complexity, that is, measuring the entropy of the fractional contribution distribution of the input element. *Sparseness* [74] evaluates complexity as sparsity, evaluating the number of non-zero coefficients in the function. This metric aligns more closely with human intuition regarding conciseness. For both metrics, a low value indicates a concise saliency map.

Finally, for localization, *Energy Pointing Game (EPG)* [83] measures the fraction of the saliency map's mass that falls within the union of the ground truth bounding boxes. A higher value signifies improved performance on localization.

3.3.4 User Reactions and Abilities. Unless stated otherwise, participants rated all items on a five-point scale ranging from 1 (strongly disagree) to 5 (strongly agree). We measured explanation satisfaction with 8 items from Hoffman et al. [35], a sample item is "the saliency maps are useful to evaluate the classifier." Trust in the saliency maps and the classifier was measured using three items respectively, using the same scale from Thielsch et al. [81], but referring to either the saliency maps or the classifier. A sample item is "I would trust the saliency maps/the classifier completely." Trustworthiness of the classifier was measured using 10 items, the item "I think the image classifier is trustworthy" was self-constructed while all other items were taken from the trustworthiness subscales ability and transparency from Höddinghaus et al. [34]. The subscale ability consists of six items, a sample item is "I think the classifier has the competence to adapt its decision to different circumstances." The subscale transparency consists of three items, a sample item is "I think the decision-making processes of the classifier are clear and transparent."

For abilities, participants in each trial were first asked to estimate the accuracy of the classifier for the respective class from 0 to 100%. We then calculated a variable that we named 'absolute accuracy difference.' We chose this name because it reflects the absolute differences between the real classification accuracy and the estimated accuracy for each trial, meaning that higher values are a stronger deviation from the real accuracy. The real classification accuracy for each class was calculated as follows: For each image, the class with the highest probability was taken as the predicted class. The number of correctly classified images, that is, whether the model prediction aligns with the human annotation in the training data, was then divided by the total number of images for each class.

Participants then indicated which of the 6 additional images of the respective class would be classified correctly or incorrectly judging from the saliency maps. For each round, we counted how many prediction errors were made by participants. Prediction errors occurred when participants predicted an image would be classified correctly by the classifier even though the classifier did not classify it correctly and vice versa. From the number of errors, we calculated participants' 'errorrate' by dividing the number of errors by the total number of predictions.

4 Results

We used linear regression, t-tests, and linear mixed models (LMM) to analyze the data. For the regression models, the three saliency map conditions (Grad-CAM vs. GBP vs. LIME) were dummy-coded with Grad-CAM as the reference group. This was done to efficiently display the main results because differences were mostly found between Grad-CAM and the other two groups. Because this method only compares Grad-CAM with LIME and Grad-CAM with GBP, we chose to additionally conduct t-tests for the comparison of GBP vs. LIME. For all ability metrics (except in the LMM analyses), that is, absolute accuracy difference and rate of correct classifications, mean values of the 12 trials for each participant were first calculated.

4.1 User Reactions

RQ1.1 asked whether the perceived explanation satisfaction is different in any of the groups. We calculated a linear regression model using explanation satisfaction as a dependent variable and the dummy-coded grouping variable as a predictor to test this hypothesis. As can be seen in Table 1, we found no significant differences. The comparison of GBP vs. LIME was conducted using a t-test and yielded no significant results, $t(103.75) = 0.37$, $p = .71$, $d = -0.07$. The response to **RQ1.1** thus is that there were no significant differences in explanation satisfaction.

RQ1.2 asked whether trust in the classifier and in the saliency map is different in any of the groups. We again calculated a linear regression of the dummy-coded group variable, this time on the trust variables, and found no significant differences between any of the groups for any of the variables (see Table 1). In the comparison of GBP with LIME, t-tests revealed no significant results for trust in the saliency map, $t(108.94) = -0.48$, $p = .63$, $d = 0.09$, trust in the classifier, $t(102.29) = -0.43$, $p = .67$, $d = 0.08$, or trustworthiness of the classifier, $t(112.48) = 0.91$, $p = .36$, $d = -0.17$. The response to **RQ1.2** thus is that there were no significant differences for trust.

4.2 User Abilities

RQ2.1 asked whether there are differences in participants' accuracy prediction relative to the true accuracy in any of the groups. Regression results revealed significant differences, with Grad-CAM resulting in more accuracy than both LIME and GBP (see Table 2). The t-test between GBP and LIME revealed no significant difference, $t(113.29) = -1.10$, $p = .27$, $d = 0.2$. The response to **RQ2.1** thus is that Grad-CAM led to better accuracy predictions than LIME and GBP.

RQ2.2 asked whether there are differences in the number of classification errors that are correctly detected in any of the groups. Regression results showed that participants in the Grad-CAM group were significantly more accurate in determining which images will be correctly or incorrectly classified than participants in the LIME group and numerically, but not significantly, better than participants in the GBP group (see Table 2). The t-test results show that there was no

Predictors	Saliency Trust			Classifier Trust		
	Estimates	CI	p	Estimates	CI	p
(Intercept)	3.35	3.10, 3.59	**<.01***	3.22	2.96, 3.48	**<.01***
Grad-CAMvsGBP	-0.01	-0.36, 0.33	.94	-0.03	-0.39, 0.34	.89
Grad-CAMvsLIME	0.06	-0.26, 0.38	.70	0.05	-0.29, 0.38	.79
Predictors	**Classifier Trustworthiness**			**Explanation Satisfaction**		
	Estimates	CI	p	Estimates	CI	p
(Intercept)	3.74	3.56, 3.92	**<.01***	3.98	3.81, 4.15	**<.01***
Grad-CAMvsGBP	-0.07	-0.33, 0.19	.60	-0.07	-0.30, 0.17	.58
Grad-CAMvsLIME	-0.18	-0.42, 0.06	.14	-0.11	-0.33, 0.11	.33

Table 1. Answering RQ1. Regression results for different models. Estimates, confidence intervals, and p-values are provided for each predictor. * indicates $p < .05$. $N = 166$

significant difference in error rates between GBP and LIME, $t(113.21) = -1.07$, $p = .28$, $d = 0.2$. The response to **RQ2.2** thus is that Grad-CAM led to less classification errors compared to LIME but not compared to GBP.

Predictors	Absolute accuracy difference			Errorrate		
	Estimates	CI	p	Estimates	CI	p
(Intercept)	18.08	16.62, 19.55	**<.01***	0.37	0.36, 0.39	**<.01***
Grad-CAMvsGBP	2.20	0.16, 4.24	**.03***	0.02	-0.01, 0.05	.12
Grad-CAMvsLIME	3.23	1.33, 5.13	**<.01***	0.03	0.01, 0.06	**<.01***

Table 2. Answering RQ2. Regression results for the abilities measures. absolute accuracy difference represents the mean absolute difference between participants accuracy estimations for each class and the classifiers' true accuracy. * indicates $p < .05$. $N = 166$

4.3 Mathematical Metrics

RQ3 asked whether there are differences in the mathematical metrics between the three approaches. We assessed the mathematical properties using the metrics discussed in Section 3.3.3. The results are shown in Table 3. Even though Grad-CAM led to better user abilities, GBP showed the most favorable values (best in four out of seven metrics) in mathematical properties. Grad-CAM only had more favorable values in two metrics (i.e., insertion and sparseness). The response to **RQ3** thus is that GBP shows the most favorable mathematical metrics.

Group	Fidelity			Complexity		Robustness	Localization
	Insertion ↑	Deletion ↓	Monotonicity ↑	Sparseness ↓	Ent.Complexity ↓	Avg. Sensitivity ↓	EPG ↑
Grad-CAM	**0.9013**	0.1273	0.7341	**0.4283**	10.4929	0.9102	0.39
GBP	0.8504	**0.0687**	**0.8299**	0.8081	**9.2976**	1.6733	**0.44**
LIME	0.8744	0.1277	0.4857	0.5564	10.1132	**0.9012**	0.41

Table 3. Answering RQ3. Mean values for the mathematical properties in the 12 trials across the different groups (Grad-CAM, GBP, LIME). An upward arrow ↑ indicates that a higher value is preferable, while a downward arrow ↓ signifies that a lower value is better. Bold values indicate the most favorable value for each property.

To assess **RQ4**, whether the mathematical properties of the saliency maps are associated with accuracy predictions in any of the saliency map groups, we used the mean of each metric for the three saliency maps in each trial. We then calculated one LMM per saliency map group for each of the seven metrics. For each LMM, we introduce a dependent variable called 'absolute accuracy difference', which is the absolute differences between participants' accuracy ratings and the true accuracy for each specific class. Mathematical properties were included as a fixed effect, and a random intercept was added for each participant to account for individual differences. These random intercepts were included because every participant provided their prediction in all twelve trials, thus the prediction of tasks was nested within participants. For example, in the model assessing the influence of the 'insertion' metric, we used a LMM with the absolute accuracy difference as the dependent variable and 'insertion' as the fixed effect. The random intercept for each participant allowed us to control for variability in baseline accuracy ratings across individuals. This approach enabled us to determine whether and to what extent the 'insertion' property had a consistent effect on abilities metrics across different participants within each saliency map group.

The LMM results can be seen in Table 4. A negative value indicates that a better score in the metric reduces the 'absolute accuracy difference,' meaning it decreases the deviation between the human prediction of accuracy and the actual accuracy based on saliency maps. Thus, if the properties that mathematical metric evaluated contribute to human accuracy prediction, we expect positive values in 'Deletion,' 'Sparseness,' 'Ent.Complexity,' and 'Avg. Sensitivity,' and negative value in 'Insertion,' 'Monotonicity,' and 'EPG'. However, our analysis brought forward mixed results.

Significant results in the expected direction were found for 'Deletion' in the GBP and LIME groups, 'Ent.Complexity' in the GBP group, and 'EPG' in the Grad-CAM and GBP group. This means that these metrics were associated with better accuracy predictions in the specific groups.

Significant results in the unexpected direction were found for 'Insertion' in the LIME group, for 'Monotonicity' and 'Ent.Complexity' in the GBP Group, for 'Sparseness' in the GBP and LIME groups, and for 'Avg. Sensitivity' in the Grad-CAM and LIME groups. This implies that these metrics in the specific groups were associated with worse accuracy predictions. The response to **RQ4** thus is that while there are associations between mathematical metrics and accuracy predictions, the picture is complex. Most of the mathematical metrics are inversely related to human understanding and only 'Deletion' and 'EPG' fit the expected positive relation.

Group	Fidelity			Complexity		Robustness	Localization
	Insertion ↑	Deletion ↓	Monotonicity ↑	Sparseness ↓	Ent.Complexity ↓	Avg. Sensitivity ↓	EPG ↑
Grad-CAM	-6.85	2.36	-5.53	13.97	**-15.26***	**-305.60***	**-13.80***
GBP	0.54	**43.43***	**32.37***	**-46.33***	**9.03***	-1.63	**-17.78***
LIME	**9.54***	**24.32***	-0.86	**-13.33***	-1.38	**-95.04***	-5.38

Table 4. Answering RQ4. Table showing linear mixed model results for the impact of different mathematical properties on the accuracy difference ratings for each saliency map approach. The analysis for each metric was conducted separately. * indicates $p < .05$. $N = 166$

5 Discussion

In this study, we aimed to assess popular saliency map approaches regarding user satisfaction, trust, abilities, and mathematical metrics. Our main results are that, even though GBP was best when it came to the overall mathematical metrics, Grad-CAM led to the best task performance, highlighting that mathematical metrics were not aligned with user abilities. Our investigation into the association between mathematical metrics and user accuracy ratings support this interpretation because all metrics except for deletion and EPG were associated with lower user understanding in specific groups, that is, worse accuracy predictions.

5.1 Contextualizing the Results

5.1.1 Performance across Evaluation Methods. All three approaches performed roughly equal regarding trust and satisfaction ratings. This is in contrast to findings by Khadivpour et al. [40], where participants preferred Grad-CAM and rated it as more understandable over LIME. This discrepancy could be caused by differences in measurement, as participants in the study by Khadivpour et al. [40] simply ranked their preference while we employed the 8 items from the explanation satisfaction scale [35] (which might be a slightly different construct as it focuses more on usefulness).

When looking at ability-related measures of understanding, our results are consistent with Morrison et al. [55], who found that Grad-CAM was more helpful for users than LIME in a similar task. Similarly, other works comparing Grad-CAM to different saliency approaches also found that Grad-CAM led to better task performance in at least one of the tests conducted, though overall results were more mixed [19, 42].

Regarding mathematical metrics, our findings are consistent with previous observations by Zhang et al. [85] and Li et al. [48] who also found that no single saliency method dominates across all metrics. Conversely, GBP showing the best performance across metrics contrasts with other studies[48, 85], where Grad-CAM performed best. This difference might arise from the dataset scope and the metrics selected. Specifically, both works choose metrics suited to their respective objectives, which include a range of metrics that favor or disadvantage certain saliency maps by design. We focus on a few widely recognized metrics rather than benchmarking or analyzing metric impacts. Furthermore, when evaluating one property using multiple metrics, we also observe disagreements between them (similar to [85]). Overall, these results make it challenging to determine which saliency map was superior based on mathematical evaluations.

5.1.2 Mathematical Metrics and User Predictions. The association of mathematical metrics to the accuracy ratings was complex and varied between metrics. For fidelity, we observed that Deletion was associated with the absolute accuracy difference as expected, but this was not the case for Insertion and Monotonicity. In fact, we found counterintuitive effects in the LIME group for Insertion and in the GBP group for Monotonicity. If the goal is to best approximate user predictions, our results suggest that Deletion might be worth weighting more in the fidelity measurement.

A low complexity is expected to aid humans in understanding saliency maps [60]. Based on the Sparseness metric, Grad-CAM is the most concise saliency map, and its Sparseness positively, but not significantly, impacted human predictions. Similarly, the Ent.Complexity metric identified GBP as the most concise, with a significant positive effect observed for GBP as well. However, because higher Sparseness was associated with less understanding in the GBP and LIME groups, the same as Ent.Complexity in the Grad-CAM group, it seems like complexity metrics are mostly not aligned with human understanding. Developers might even have to choose between optimizing for metrics of complexity or for human understanding, as our results imply choosing one could negatively impact the other.

In terms of robustness, higher Sensitivity is associated with greater accuracy, suggesting that humans tend to benefit from more sensitive saliency map methods. This is contrary to our expectations, as we predicted less sensitive maps to deliver more reliable insight into the functioning of the classifier. One explanation for this could be that higher Sensitivity enables humans to better assess how well a saliency map can react to even the smallest changes.

EPG, as a metric for evaluating localization, measures the overlap between the saliency map and the object annotated by humans using a bounding box, aligning with human annotation of objects by concept. This alignment could be one explanation for the positive relation between EPG and human understanding. Of the metrics we examined, EPG seems to be the one most closely associated with user accuracy predictions, at least in the GBP and Grad-CAM groups. This means that localization might be the most useful to serve as a proxy for user metrics.

These results provide more nuance to the often made hypothesis that mathematical metrics can serve as a proxy for human studies [16, 23]. Not only did the different approaches fare differently across the saliency map methods, but the connection between the mathematical metrics and the accuracy rating abilities measure were also rather mixed. According to our results, only EPG and deletion could serve to gain an idea of how users might fare in terms of abilities and even then, only for specific saliency map approaches. Interestingly, this finding is in contrast to findings by Colin et al. [19], who found that EPG was not associated with indicating user performance. As things stand, this means that the mathematical metrics that we used in the current study cannot be used as proxies for findings of user studies.

5.2 Evaluating XAI Approaches

The finding that the evaluation methods used in our study diverged in their assessment raises fundamental questions about the alignment between mathematical and user evaluations in XAI. While mathematical metrics are crucial to ensure that explanations are technically sound and accurately reflect the inner workings of a model, our results shows that they cannot be easily associated with user abilities. This discrepancy suggests that focusing solely on mathematical metrics may overlook critical aspects of how users interact with and benefit from explanations. Instead, our findings underscore the need for a holistic assessment of user and mathematical metrics when choosing XAI approaches. For saliency maps, for example, this could be a structured method of classification such as saliency cards [13].

Furthermore, the trust and satisfaction ratings of users did not provide much insight into the meaningfulness and usefulness of an explainability approach for user abilities, and thus their actual understanding of a classifier. This aligns with findings in cognitive science (e.g., [72]), demonstrating that perceived understanding often diverges from actual understanding. Such subjective measures can be influenced by cognitive biases, user familiarity with the explanation style, or even the aesthetic appeal of an XAI approach's outputs and might not translate into increased understanding and the factual correctness of the approach [11]. Examining user abilities and mathematical properties are important complements to mitigate such risks, especially as our study clearly shows that human reactions, user abilities, and mathematical metrics can point in different directions.

5.3 Practical Implications

Our study does not identify a best saliency map method, suggesting that the optimal choice depends on the specific context. If users' abilities and objective understanding are a priority, Grad-CAM appears to be the most suitable option. Both our findings and prior literature (e.g., [55]) indicate that Grad-CAM helps users understand classifiers. Additionally, our results demonstrate that it fosters a moderate level of trust and a high degree of explanation satisfaction.

On the other hand, in contexts where faithfulness to the underlying model is paramount—such as regulatory audits or safety-critical applications—our findings suggest that GBP may be a more appropriate choice. However, as some studies have also identified Grad-CAM as superior in this regard [48], the evidence remains inconclusive, and further research is needed to provide a definitive recommendation.

Our results are also in line with research suggesting that saliency maps may have limited usefulness in helping participants make accurate classification predictions (e.g., [5]). In our experiment, participants' average accuracy predictions deviated by ±18-21% from the actual accuracy, and the classification prediction error rate was 37-40%, only slightly above chance level. This is particularly noteworthy because we only examined participants with a background in

Information and Communication Technology (ICT) in our study. As these participants likely have a better understanding of saliency maps than laypeople, objective measures of user understanding might be even worse for a broader audience.

5.4 Limitations and Future Work

One limitation of our study is that we only had English-speaking participants with a background in ICT. This limits the sample in terms of culture and expertise. In particular, it is likely that lay people who are confronted with saliency maps have a different attitude towards these explainability methods. However, we deliberately chose these participants, as they are the most likely to encounter and use saliency maps in their professional work, and thus possess the necessary expertise to critically evaluate them. Future studies should follow a similar procedure with other stakeholder groups.

Lastly, we presented the Grad-CAM saliency maps with the original image clearly visible in the background, whereas for LIME, the original image was only faintly visible, and for GBP, no background image was provided. This is a common presentation for each of these approaches [2, 40, 41, 76]. Nonetheless, having the original image in the background of the saliency map may assist users by reducing the cognitive demands to mentally overlay the original with the saliency map to identify which features contributed to the classification. This could partly explain why participants in the Grad-CAM group demonstrated the highest user abilities compared to other approaches, though the finding is still consistent with prior research (e.g., [19, 42, 55]). To minimize the effects of this discrepancy, we displayed the original image directly above the corresponding saliency map for all groups. Results also showed that users did not have a preference for any of the approaches with regards to trust or explanation satisfaction which might indicate that the effect of the background image might be small. Even so, if there is an effect, ensuring that saliency maps minimize cognitive demands could be an important design consideration for future research. Future studies could explore whether overlapping saliency maps with the original images enhances user understanding compared to side by side presentation.

6 Conclusion

This study explored the evaluation of saliency maps in XAI, comparing Grad-CAM, LIME, and GBP across various evaluation methods. Our findings revealed that while trust and satisfaction ratings did not significantly differ across the methods, Grad-CAM demonstrated that it can better help users to predict model performance and correct classifications. However, GBP was better evaluated with respect to most mathematical metrics. All metrics except for deletion and EPG were also associated with a lower user understanding in specific groups. These results underscore the importance of a critical question in XAI: are mathematical metrics and user-centered evaluations aligned? While technically sound explanations are often essential, they do not necessarily improve user interaction or decision-making. Our study shows that most mathematical metrics are likely not suited to serve as proxies for user studies but that the two could serve to complement each other in important ways. Future research could aim to explore how to match mathematical and user metrics to best meet context-dependent desiderata of specific XAI approaches.

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A Survey Material

A.1 Introduction

Welcome to the Study!

Thank you for being interested in our experiment. This research is *[redacted for review]*.

In this experiment, you will evaluate an image classifier based on AI. The classifier uses a visualization technique called "saliency map". Don't worry if you have not heard of this before, all important terms will be explained at the beginning of the study.

Please only take part in this study if you are over the age of 18, proficient with the English language and on a PC, laptop or (large) tablet. The experiment takes around 40 minutes.

(1) Please insert your Prolific ID here *[free text]*

A.2 Data Protection

Consent according to the EU General Data Protection Regulation (GDPR)

1. What personal data do we collect and process?

We take the protection of your personal data very seriously. We treat your personal data confidentially and in accordance with the statutory data protection regulations and this privacy policy. Participation in the study is anonymous, and no conclusions can be drawn about your identity under any circumstances.

We use your personal data for conducting our study in compliance with the applicable data protection provisions. Below, we will explain which personal data is collected and stored. You will also receive information on how your data is used and what rights you have regarding the use of your data.

The following personal data is collected when participating in the study:

- Gender
- Age
- Education
- Experience with Saliency Maps

This data is collected exclusively for scientific purposes. By signing this document, you permit *[redacted for review]* to use the information you provide in the survey for these scientific purposes.

2. Information about the rights of data subjects
You have the right to:

- Request information about your personal data processed by us in accordance with Art. 15 GDPR. In particular, you can request information about the processing purposes, the category of personal data, the categories of recipients to whom your data has been or will be disclosed, the planned storage duration, the existence of a right to rectification, deletion, restriction of processing or objection, the existence of a right to complain, the origin of your data if it was not collected by us, and the existence of automated decision-making including profiling and, if applicable, meaningful information about its details.
- Request the correction of incorrect or incomplete personal data stored by us without delay in accordance with Art. 16 GDPR.
- Request the deletion of your personal data stored by us in accordance with Art. 17 GDPR, unless the processing is necessary for exercising the right to freedom of expression and information, to fulfill a legal obligation, for reasons of public interest, or to assert, exercise, or defend legal claims.
- Request the restriction of processing of your personal data in accordance with Art. 18 GDPR, as far as the accuracy of the data is disputed by you, the processing is unlawful, but you oppose its deletion, and we no longer need the data, but you require it to assert, exercise, or defend legal claims or you have objected to the processing in accordance with Art. 21 GDPR.
- Receive your personal data that you have provided to us in a structured, commonly used, and machine-readable format or to request the transfer to another controller in accordance with Art. 20 GDPR.
- Withdraw your consent at any time in accordance with Art. 7 para. 3 GDPR. This means that we will no longer continue the data processing based on this consent in the future.
- Complain to a supervisory authority in accordance with Art. 77 GDPR. As a rule, you can contact the supervisory authority of your usual place of residence or workplace or our office.

3. Information about the right to object
If your personal data is processed based on legitimate interests pursuant to Art. 6 para. 1 sentence 1 lit. f GDPR, you have the right to object to the processing of your personal data in accordance with Art. 21 GDPR, provided that there are reasons arising from your particular situation or the objection is directed against direct marketing. In the latter case, you have a general right to object that we will implement without the need to specify a particular situation.

If you wish to exercise your right of withdrawal, an email to *[redacted for review]* is sufficient.

The supervisory authority responsible for *[redacted for review]* is *[redacted for review]*

Data retention period: Seven years.

(2) Do you consent to these terms? ○ Yes, I consent. ○ No, I do not consent (end experiment)

Please read the explanations on the following pages carefully. You will be asked to answer comprehension questions later, and you cannot take part in the study if you answer incorrectly twice in a row.

If you are already proficient with the concept of evaluating AI image classifiers using saliency maps and confident you can answer comprehension questions, you may skip the explanations.

A.3 Background Information

Please read the explanations on the following pages carefully. You will be asked to answer comprehension questions later, and you cannot take part in the study if you answer incorrectly twice in a row.

If you are already proficient with the concept of evaluating AI image classifiers using saliency maps and confident you can answer comprehension questions, you may skip the explanations.

What are Saliency Maps

In this study, we will be exploring something called "saliency maps." To help you understand what these are, let's start with a simple explanation.

Imagine you are looking at a picture. Your eyes and brain naturally focus on certain parts of the image more than others. For example, you might notice a bright color, a face, or a moving object first. This is because those parts of the picture stand out to you—they are more "salient."

A saliency map is a visual tool that shows which parts of an image are most likely to catch your attention. It highlights the areas that are most prominent or important in a picture. These maps can be created by computers using different techniques to predict where people will look when they see an image.

Here's how it works:

- **Image Analysis:** A computer program analyzes an image to find the features that stand out, like colors, edges, or unique shapes.
- **Highlighting Salient Areas:** The program then creates a map that highlights these features. The brighter or more colorful areas on the map show where your attention is most likely to go.

Saliency maps are not only helpful to understand what features stand out to humans but also particularly useful in understanding how computers learn to recognize images. When a computer is trained to identify objects in pictures, it looks at many examples and learns to focus on the important parts that help it make a correct identification. **The saliency map shows us which parts of the image the computer thinks are important.**

In our study, we will show you different saliency maps and measure how accurate you think the AI classification is.

(3) ○ I have read and understood the explanation

A.3.1 Grad-CAM.

In this study, you will see Saliency Maps based on the Grad-CAM approach

How to Interpret Saliency Maps Generated Using the Grad-CAM Approach

Saliency maps generated using the Grad-CAM (Gradient-weighted Class Activation Mapping) approach are visual tools that help us understand which parts of an image a neural network (like a deep learning classifier) focuses on to make its decision. Here's a simple explanation to interpret these maps:

What is Grad-CAM?

Grad-CAM is a technique used to explain the predictions of AI image classifiers. It produces a heatmap that shows the areas of the image that are most influential.
Reading a Saliency Map:

- **Colors:** The heatmap uses colors to indicate the importance of different regions. Typically, warmer colors (reds, yellows) represent high importance, and cooler colors (blues, greens) represent low importance.

Interpreting the Map:

- **High Importance Areas:** Look for the regions highlighted in warm colors. These are the parts of the image the model is using most to make its classification decision. For example, if the task is to classify an image of a dog, and the dog's face is highlighted in red, it indicates that the model considers the face important for making its decision.
- **Low Importance Areas:** Regions in cooler colors are considered less important by the model for the classification task. If the background or non-essential parts of the animal are highlighted in blue or green, it means these areas have little influence on the classification.

Assessing Classifier Performance:

- **Correct Focus:** If the high importance areas correspond to the distinctive features of the animal (e.g., the face, body shape, distinctive markings or color), it suggests that the classifier is likely working well and making decisions based on relevant features.
- **Incorrect Focus:** If the high importance areas are unrelated to the animal, it might indicate that the classifier is not focusing on the correct parts of the image, which could lead to poor performance.

Example
Imagine you have an image of a cat, and the saliency map generated by Grad-CAM shows the following:

- **High Importance (Red/Yellow):** The map highlights the cat's face and ears.
- **Low Importance (Blue/Green):** The map shows cooler colors around the background and the less distinctive parts of the body.

Interpretation: This suggests the classifier is correctly focusing on the key features that define a cat, such as the face, whiskers and ears, which are typically important for recognizing cats. Therefore, the classifier is likely working well for this image.

By looking at the saliency map, you can visually infer whether the model's attention aligns with human intuition, providing insight into how well the classifier is performing its task.

Here is an example of a Saliency Map generated using Grad-CAM (left). [See Figure 1a]

Please adjust the brightness of your monitor if you have difficulties seeing the Saliency Map.

(4) ○ I have read and understood the explanation

A.3.2 Guided Backpropagation.

In this study, you will see Saliency Maps based on the Guided Backpropagation approach
How to Interpret Saliency Maps Generated Using the Guided Backpropagation Approach
Saliency maps generated using the Guided Backpropagation approach are visual tools that help us understand which parts of an image a neural network (like a deep learning classifier) focuses on to make its decision. Here's a simple explanation to interpret these maps:

What is Guided Backpropagation?
Guided Backpropagation is a technique used to explain the predictions of AI image classifiers. It creates a detailed map that highlights the specific features of the image that are most influential.
Reading a Saliency Map:

- **Colors:** The saliency map typically uses grayscale values to indicate the importance of different regions. Brighter areas represent high importance, while darker areas represent low importance.

Interpreting the Map:

- **High Importance Areas:** Look for the regions highlighted in bright colors. These are the parts of the image the model is using most to make its classification decision. For example, if the task is to classify an image of a dog, and the dog's face is highlighted in bright white, it indicates that the model considers the face important for making its decision.
- **Low Importance Areas:** Darker regions are considered less important by the model for the classification task. If the background or non-essential parts of the animal are darker, it means these areas have little influence on the classification.

Assessing Classifier Performance:

- **Correct Focus:** If the high importance areas correspond to the distinctive features of the animal (e.g., the face, body shape, distinctive markings or color), it suggests that the classifier is likely working well and making decisions based on relevant features.
- **Incorrect Focus:** If the high importance areas are unrelated to the animal, it might indicate that the classifier is not focusing on the correct parts of the image, which could lead to poor performance.

Example
Imagine you have an image of a cat, and the saliency map generated by Guided Backpropagation shows the following:

- **High Importance (Bright Areas):** The map highlights the cat's face and ears.
- **Low Importance (Dark Areas):** The map shows darker areas around the background and the less distinctive parts of the body.

Interpretation: This suggests the classifier is correctly focusing on the key features that define a cat, such as the face, whiskers and ears, which are typically important for recognizing cats. Therefore, the classifier is likely working well for this image.

By looking at the saliency map, you can visually infer whether the model's attention aligns with human intuition, providing insight into how well the classifier is performing its task.

Here is an example of a Saliency Map generated using Guided Backpropagation (left). [See Figure 1b]

Please adjust the brightness of your monitor if you have difficulties seeing the Saliency Map.

(5) ○ I have read and understood the explanation

A.3.3 LIME.

In this study, you will see Saliency Maps based on the LIME approach

How to Interpret Saliency Maps Generated Using the LIME Approach

Saliency maps generated using the LIME (Local Interpretable Model-agnostic Explanations) approach are visual tools that help us understand which parts of an image a neural network (or any classifier) focuses on to make its decision. Here's a simple explanation to interpret these maps:
What is LIME?
LIME is a technique used to explain the predictions of AI image classifiers. LIME highlights which parts of the image are most influential.
Reading a Saliency Map:

- **Colors:** The saliency map typically uses highlighted segments to indicate importance. Segments that are important for the classifier's decision are highlighted, while less important segments are not.

Interpreting the Map:

- **High Importance Areas:** Look for the regions that are highlighted. These are the parts of the image the model is using most to make its classification decision. For example, if the task is to classify an image of a dog, and the dog's face is highlighted, it indicates that the model considers the face important for making its decision.
- **Low Importance Areas:** Non-highlighted regions are considered less important by the model for the classification task. If the background or non-essential parts of the animal are not highlighted, it means these areas have little influence on the classification.

Assessing Classifier Performance:

- **Correct Focus:** If the high importance areas correspond to the distinctive features of the animal (e.g., the face, body shape, distinctive markings or color), it suggests that the classifier is likely working well and making decisions based on relevant features.
- **Incorrect Focus:** If the high importance areas are unrelated to the animal, it might indicate that the classifier is not focusing on the correct parts of the image, which could lead to poor performance.

Example
Imagine you have an image of a cat, and the saliency map generated by LIME shows the following:

- **High Importance (Bright Areas):** The map highlights the cat's face and ears.
- **Low Importance (Dark Areas):** The map shows darker areas around the background and the less distinctive parts of the body.

Interpretation: This suggests the classifier is correctly focusing on the key features that define a cat, such as the face, whiskers and ears, which are typically important for recognizing cats. Therefore, the classifier is likely working well for this image.

By looking at the saliency map, you can visually infer whether the model's attention aligns with human intuition, providing insight into how well the classifier is performing its task.

Here is an example of a Saliency Map generated using LIME (left). [See Figure 1b]

Please adjust the brightness of your monitor if you have difficulties seeing the Saliency Map.

(6) ○ I have read and understood the explanation

A.3.4 Comprehension Questions.

Please answer the following questions before starting the experiment

It is very important that you understood the explanations regarding saliency maps. Please answer the following questions.

You can use the “back” button at the end of the page to re-read the explanations the questions are based on.

If you answer incorrectly, you will jump back in the questionnaire and see the explanations again. If you answer incorrectly twice in a row, you will be asked to return your submission on Prolific and the questionnaire will end.

(7) **Based on the explanation provided, which of the following best describes the purpose of a saliency map in the context of this study? (Only one is correct)** ○ A saliency map is a tool used to edit images by enhancing their colors and shapes. ○ A saliency map is a visual representation that shows which parts of an image are most important for an AI-based classification. ○ A saliency map highlights the least important parts of an image. ○ A saliency map is a method to measure how quickly an AI can process an image.

(8) **Which of the following statements best explains how to interpret high importance areas on a saliency map generated using Guided Backpropagation? (Only one is correct)** ○ High importance areas are the darkest regions of the map, indicating the features least used by the model for classification. ○ High importance areas are the brightest regions of the map, indicating the features most used by the model for classification. ○ High importance areas are highlighted in color to show the most aesthetically pleasing parts of the image. ○ High importance areas are blurred to indicate where the model is uncertain about the classification.

(9) **What does it suggest about a classifier’s performance if the saliency map highlights the background of an image as highly important? (Only one is correct)** ○ The classifier is performing well and correctly focusing on the relevant features of the image. ○ The classifier is not performing well and is focusing on irrelevant features of the image. ○ The classifier is indicating areas that need more training data. ○ The classifier is highlighting the most aesthetically pleasing parts of the image.

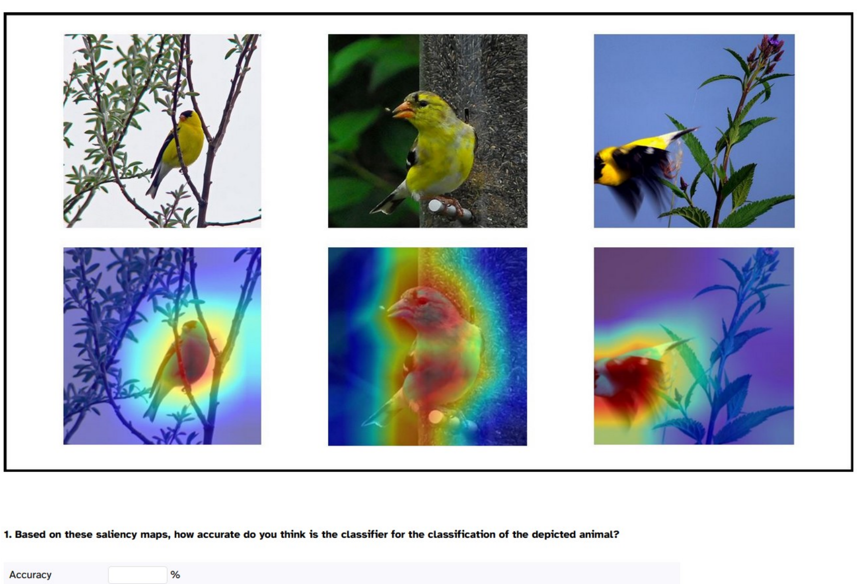


Fig. 2. Example of the Grad-CAM task

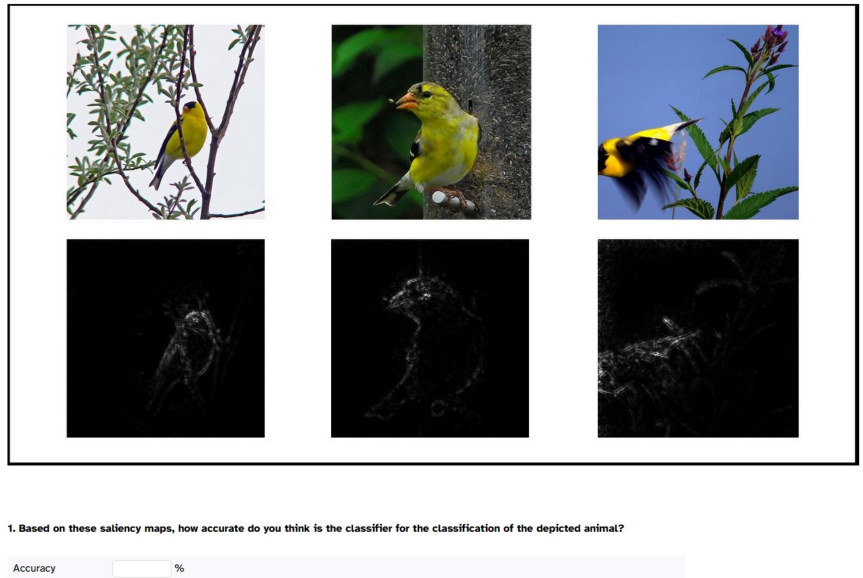


Fig. 3. Example of the GBP task

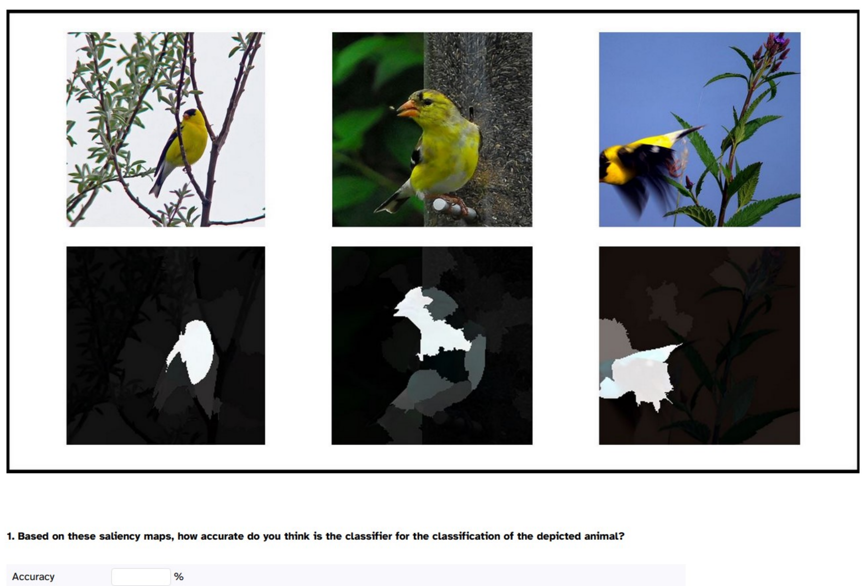
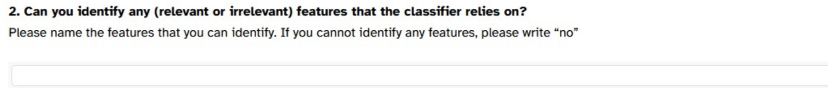


Fig. 4. Example of the LIME task

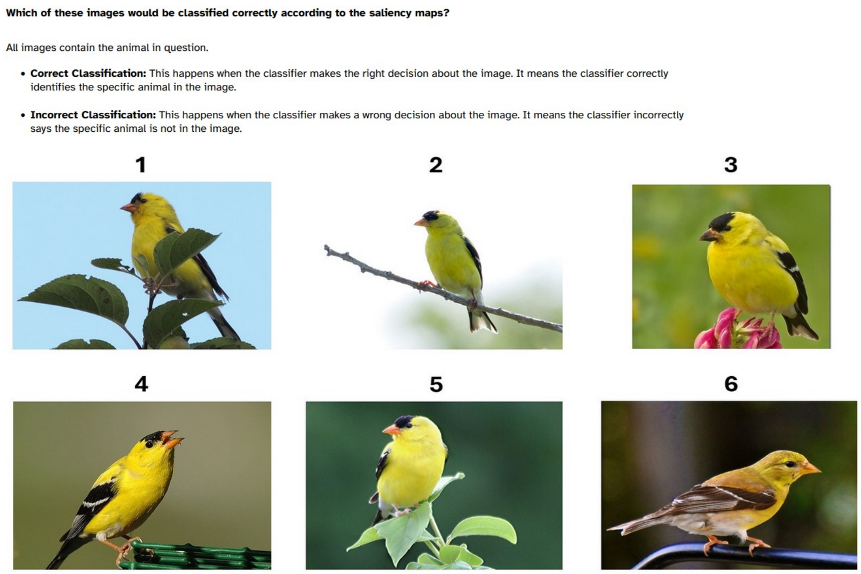
A.3.5 User task.



2. Can you identify any (relevant or irrelevant) features that the classifier relies on?

Please name the features that you can identify. If you cannot identify any features, please write “no”

Fig. 5. Continuation of the user task (identical in all groups)

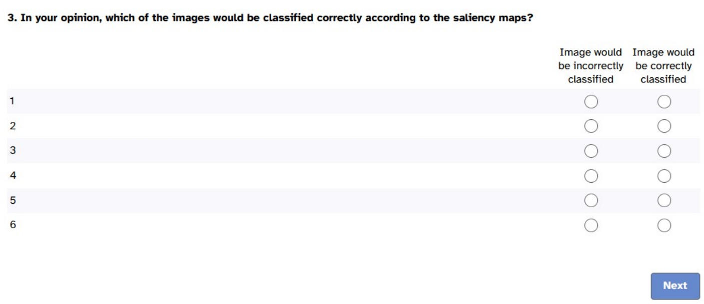


Which of these images would be classified correctly according to the saliency maps?

All images contain the animal in question.

- **Correct Classification:** This happens when the classifier makes the right decision about the image. It means the classifier correctly identifies the specific animal in the image.
- **Incorrect Classification:** This happens when the classifier makes a wrong decision about the image. It means the classifier incorrectly says the specific animal is not in the image.

Fig. 6. Continuation of the user task (identical in all groups)



3. In your opinion, which of the images would be classified correctly according to the saliency maps?

	Image would be incorrectly classified	Image would be correctly classified
1	○	○
2	○	○
3	○	○
4	○	○
5	○	○
6	○	○

Next

Fig. 7. Continuation of the user task (identical in all groups)